Cryptographic Hash Functions

Hash Functions

Cryptographic hash functions are mathematical algorithms that transform input data into a fixed-length sequence of characters, referred to as a hash value. Cryptographic hash functions are intended to be fast, deterministic, and one-way, meaning that even a minor change in input yields a very different hash.

Properties of Cryptographic Hash Functions

Deterministic: The same input always generates the exact same hash output, ensuring consistent and reliable verification of data.

Fast Computation: Cryptographic hash functions are designed to process inputs quickly and efficiently, making them practical for handling large datasets and real-time applications.

Properties of Cryptographic Hash Functions

Pre-image Resistance: It is computationally infeasible to reverse-engineer or retrieve the original input data from its hash value, protecting sensitive information from exposure.



Second Pre-image Resistance: Given an input and its hash, it is extremely difficult to find a different input that produces the same hash, preventing impersonation or forgery.

Collision Resistance: The function minimizes the chance that two distinct inputs will produce identical hash values, ensuring unique data fingerprints for security and integrity.

Avalanche Effect: Even a tiny change in the input, such as flipping a single bit, causes a significant and unpredictable change in the hash output, enhancing the function's sensitivity to data modifications.

Popular Cryptographic Hash Algorithms

MD5 (Message Digest Algorithm 5)

Once widely used for data integrity and digital signatures, MD5 is now considered insecure due to vulnerabilities that allow attackers to generate hash collisions easily. Its speed and simplicity made it popular historically, but it is no longer recommended for security-critical applications.

SHA-1 (Secure Hash Algorithm 1)

SHA-1 improved upon MD5 with a longer hash length and better resistance to collisions. However, advances in computational power and cryptanalysis exposed weaknesses, leading to practical collision attacks. Consequently, SHA-1 is deprecated for most security uses, including SSL/TLS certificates and digital signatures.

Popular Cryptographic Hash Algorithms

SHA-2 Family (SHA-256, SHA-512)

The SHA-2 family is currently the industry standard for cryptographic hashing, offering robust security with longer hash outputs of 256 and 512 bits. These algorithms provide strong collision and pre-image resistance, making them the preferred choice for secure communication protocols, blockchain technologies, and password hashing.

SHA-3 (Keccak)

Adopted as the latest NIST standard, SHA-3 uses a unique sponge construction different from SHA-2, enhancing security and flexibility. It offers comparable hash lengths with improved resistance to certain types of attacks, making it suitable for applications demanding long-term security.

HOW HASH FUNCTION WORK?

HASH ALGORITHM

- Hash function or hash algorithm creates a unique digital fingerprint of data.
- The digital fingerprint of data is called digest, or message digest or simply hash.
- Hash algorithm is primarily used for comparison purpose, nor for encryption.



Or digest

CIA Triad-Confidentiality-Integrity-Availability

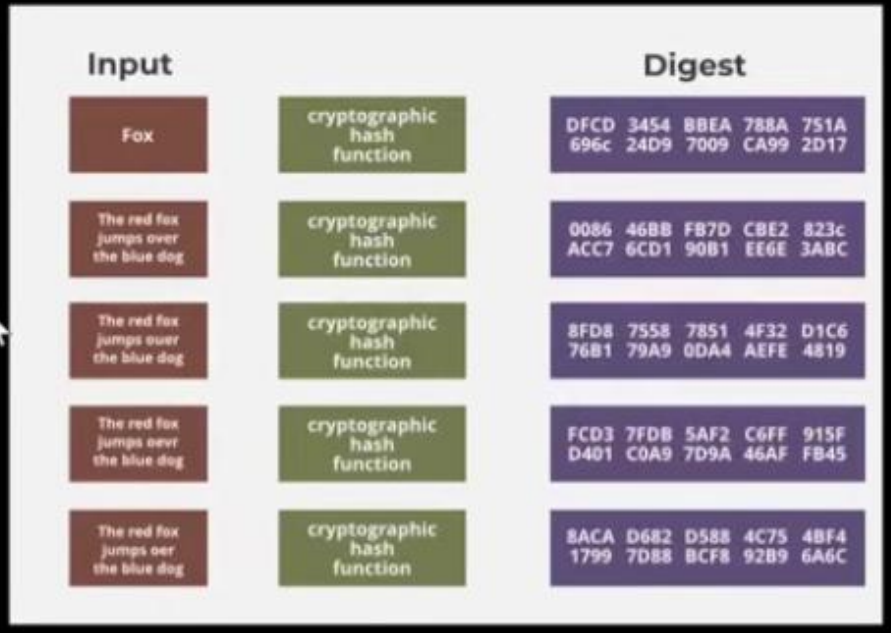
Confidentiality (C): Ensuring that information is not disclosed to unauthorized individuals, entities, or processes. Only the intended recipient should be able to read the data. Like password to login, other cannot.

Integrity (I): Ensuring that data is accurate, complete, and has not been altered or tampered with, either accidentally or maliciously, while in storage or transit.

vailability (A): Ensuring that information and systems are accessible to authorized users whenever they are needed.

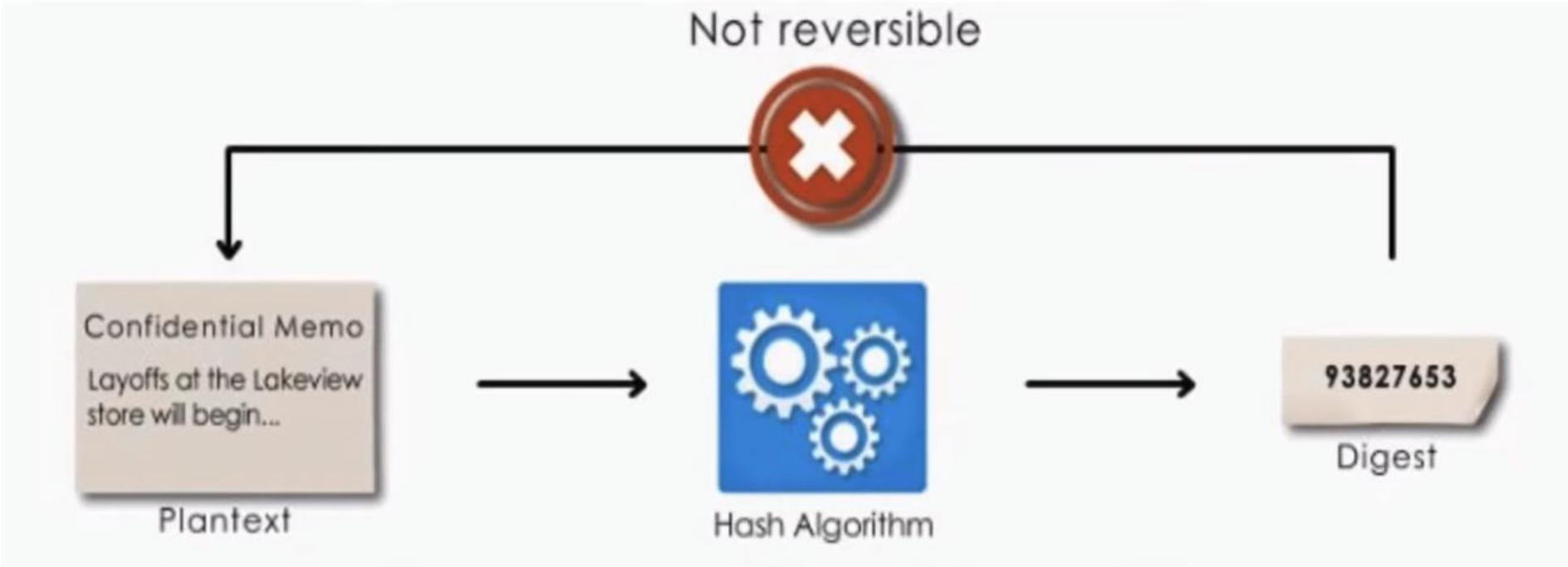
Integrity

Let's assume Host 'A' wants to send data to Host 'B' to maintain integrity. A hash function will run over the data and produce an arbitrary hash value **H1** which is then attached to the data. When Host 'B' receives the packet, it runs the same hash function over the data which gives a hash value of **H2**. Now, if **H1** = **H2**, this means that the data's integrity has been maintained and the contents were not modified.



HASH ALGORITHM

- Many secure Hash Algorithm have three basic characteristics:
  - ❑1.Secure
  - ❑2.Fixed size
  - ❑3.Unique



3 BASIC CHARACTERISTICS

- Secure-: Non-reversible function

Example- Hash Algorithm: MD5

Original Data	Digest	digest size
a	0cc175b9c0f1b6a831c399e269772661	128 bit
ab	187ef4436122d1cc2f40dc2b92f0eba0	128 bit
abc	900150983cd24fb0d6963f7d28e17f72	128 bit
abc123	e99a18c428cb38d5f260853678922e03	128 bit
abc12345	d6b0ab7f1c8ab8f514db9a6d85de160a	128 bit
abcdefg12345	6e97687128e8b80da01a5400813ae4ad	128 bit
abcdefghijklmn1234567890	3d4f26c33ab8c83f1edbdec1bf92b904	128 bit
abcdefghijklmnopqrstuvwxyz1234567890	7f7d52a001b9d2c71b6bae1f189f41f3	128 bit

- **Fixed size-: Short of long data will produce the fixed-size digest.**

3 BASIC CHARACTERISTICS

Example- Hash Algorithm: MD5

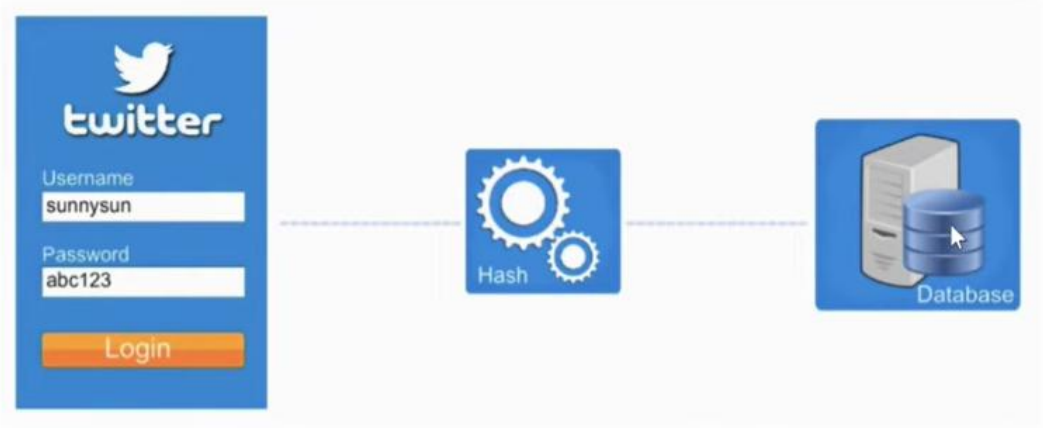
Original Data	Digest	digest size
a	0cc175b9c0f1b6a831c399e269772661	128 bit
ab	187ef4436122d1cc2f40dc2b92f0eba0	128 bit
abc	900150983cd24fb0d6963f7d28e17f72	128 bit
abc123	e99a18c428cb38d5f260853678922e03	128 bit
abc12345	d6b0ab7f1c8ab8f514db9a6d85de160a	128 bit
abcdefg12345	6e97687128e8b80da01a5400813ae4ad	128 bit
abcdefghijklmn1234567890	3d4f26c33ab8c83f1edbdec1bf92b904	128 bit
abcdefghijklmnopqrstuvwxyz1234567890	7f7d52a001b9d2c71b6bae1f189f41f3	128 bit

- **Unique-: Two different data sets cannot produce the same digest.**

3 BASIC CHARACTERISTICS

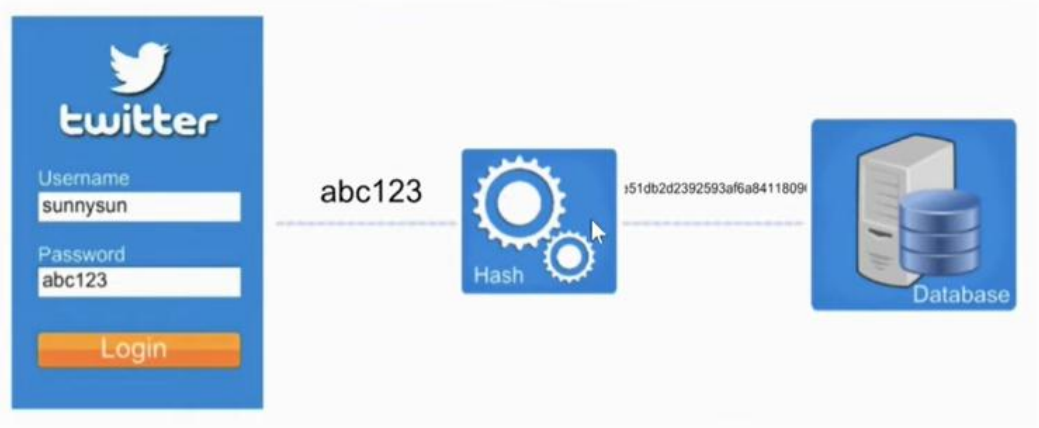
HASH ALGORITHM

- Hashing function/algorithm is commonly used for passwords.



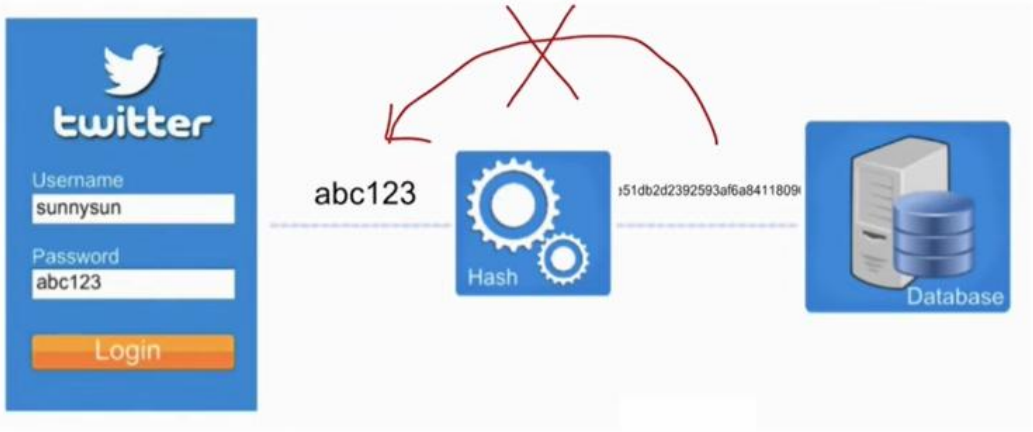
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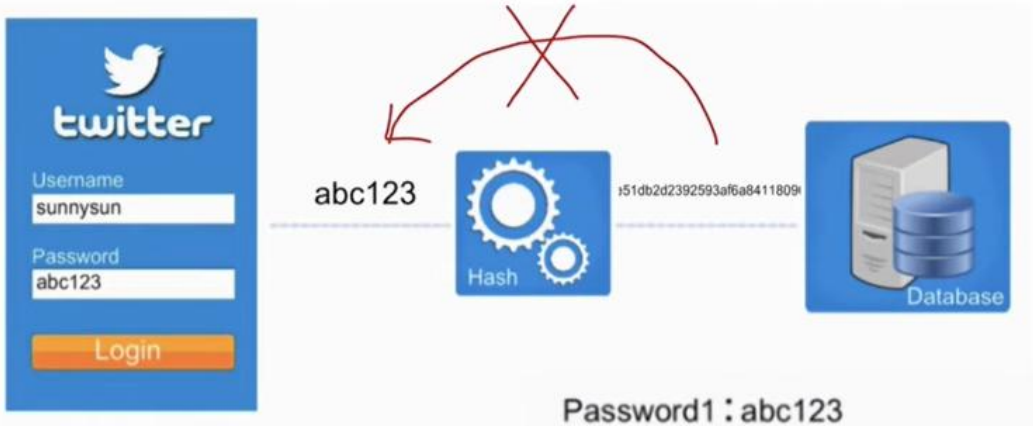
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HASH ALGORITHM

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SHA-256 (Secure Hash Algorithm 256)

Password1：abc123

6ca13d52ca70c883e0f0bb101e425a89e8624de51db2d2392593af6a84118090

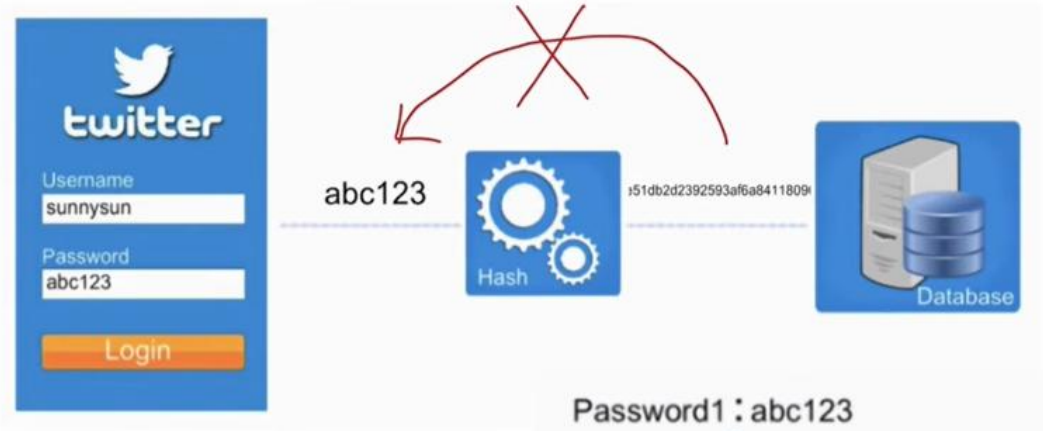
256-bit

New Password: Good morning

256 bits 90a90a48e23dcc51ad4a821a301e3440ffeb5e986bd69d7bf347a2ba2da23bd3

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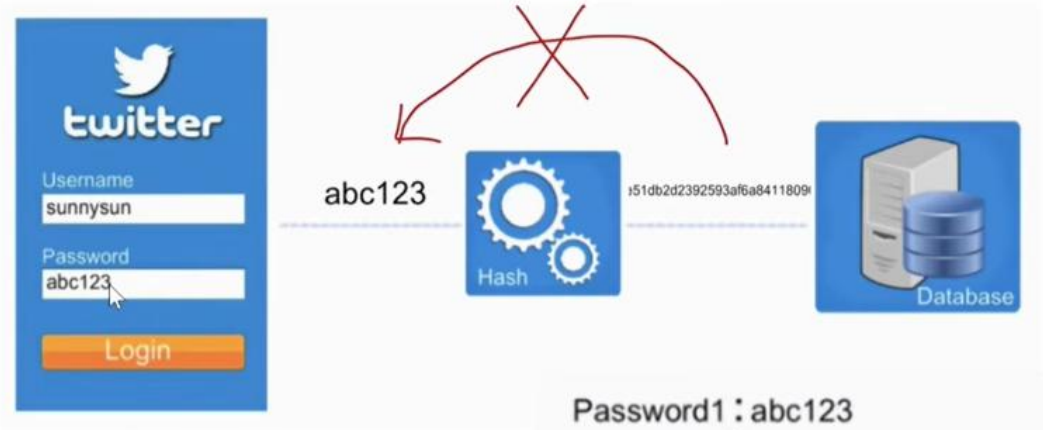
- All passwords in twitter or any secure database should be stored in the form of digests.
- Every time you login, your password is hashed into a digest, which then compared against the one stored in the database.
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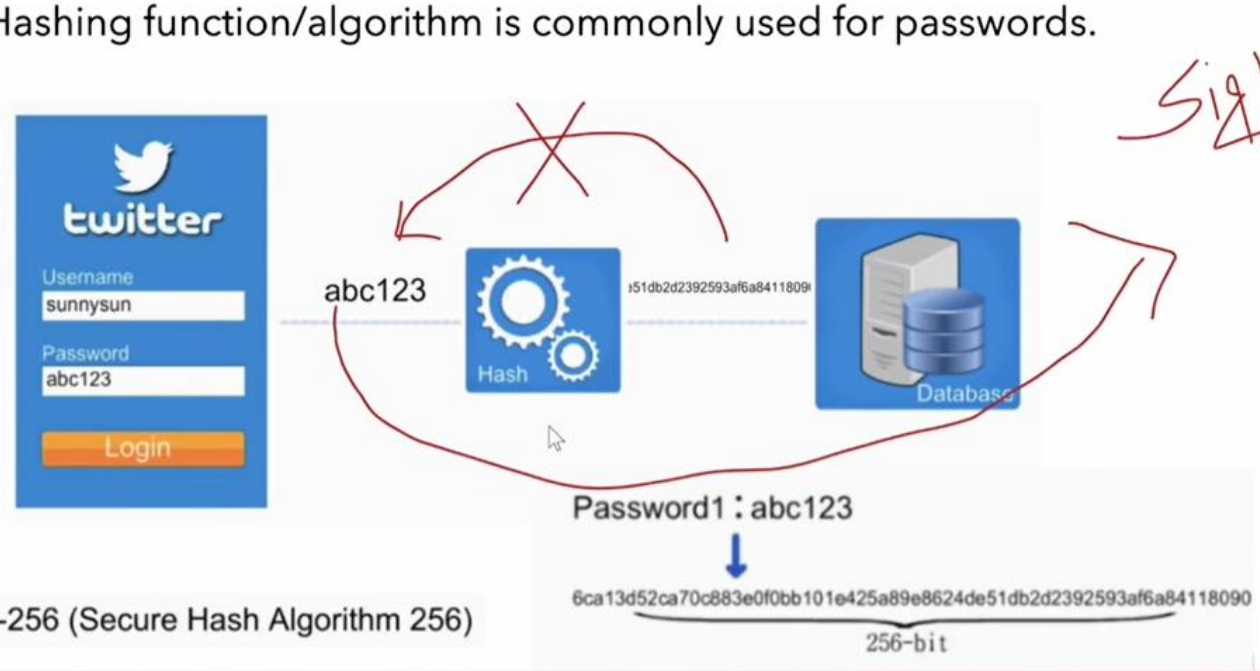
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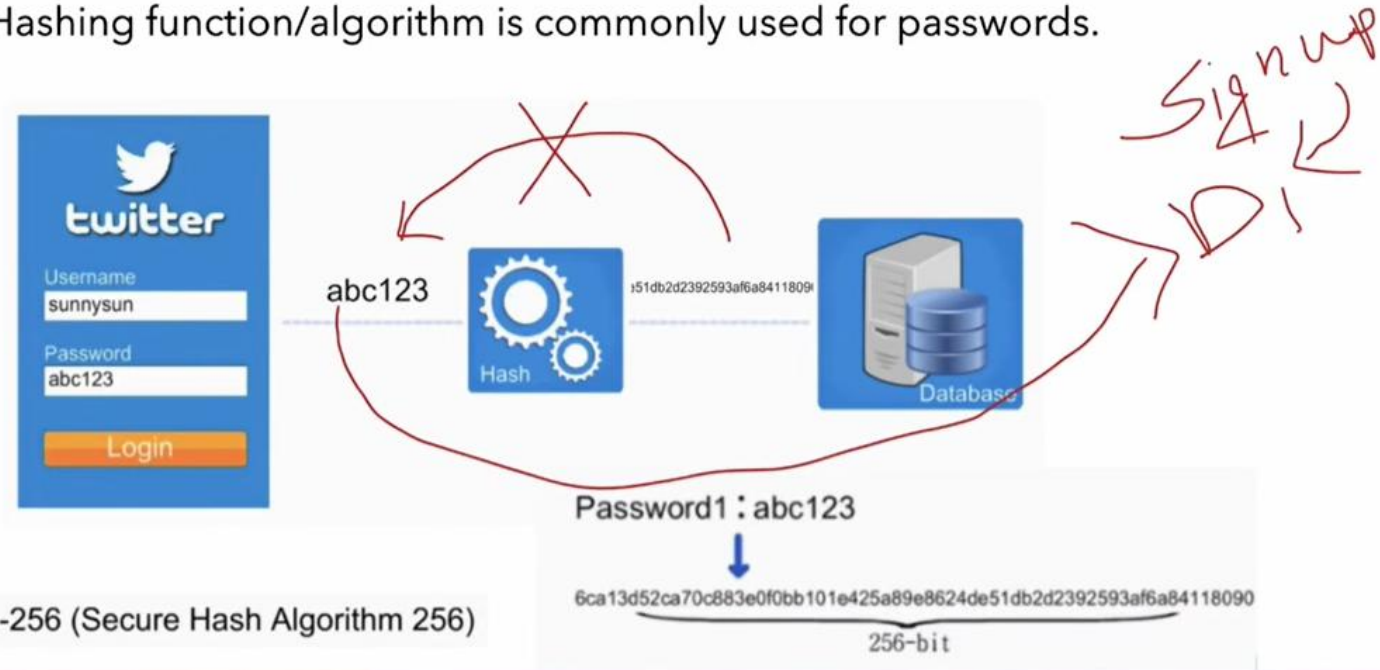
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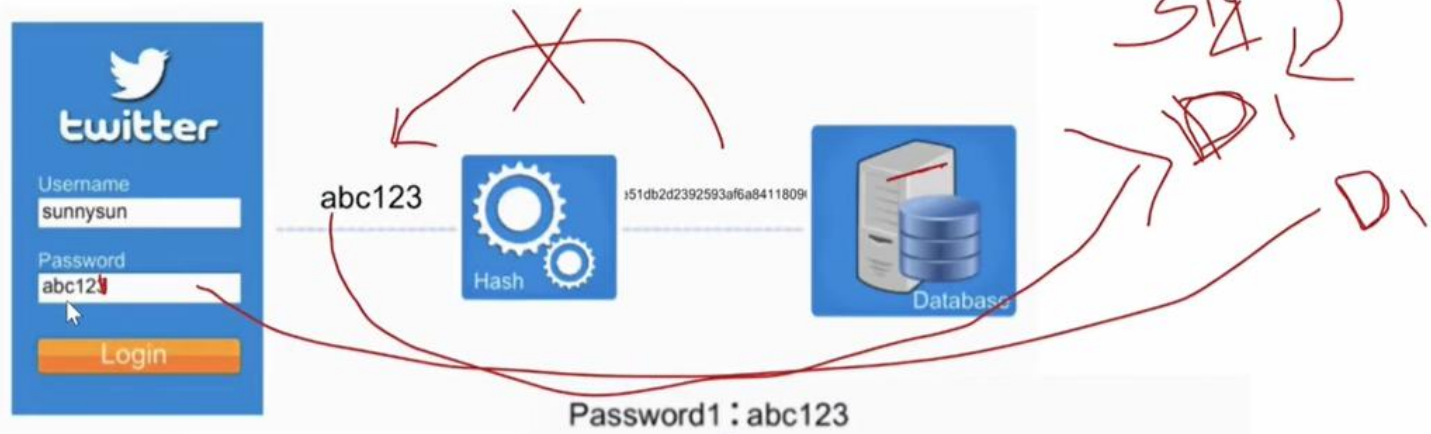
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$D_1 = D_1$

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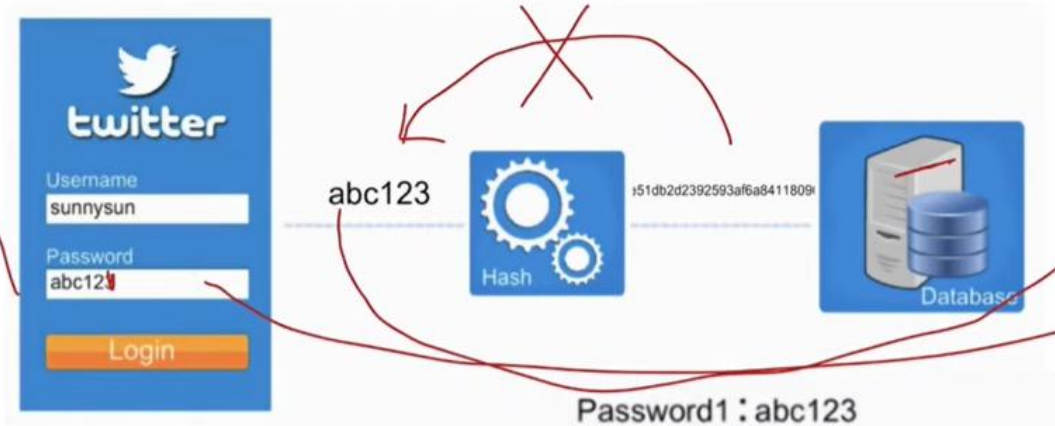
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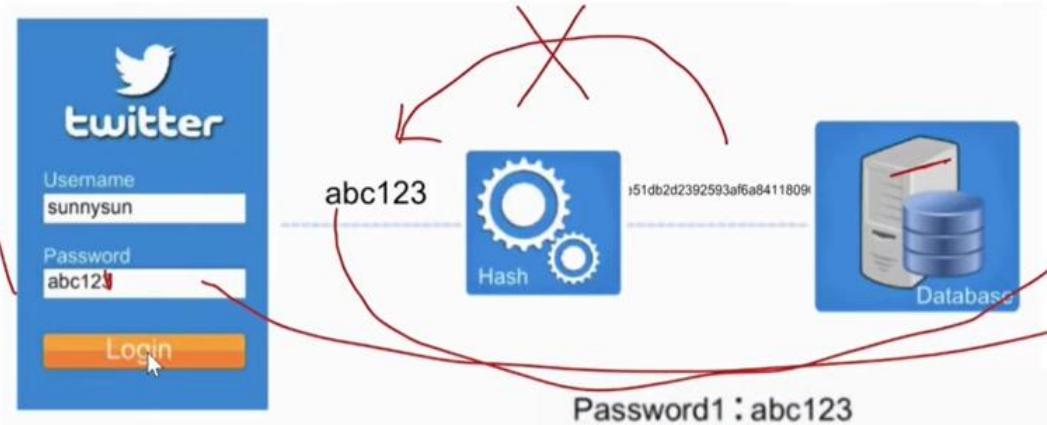
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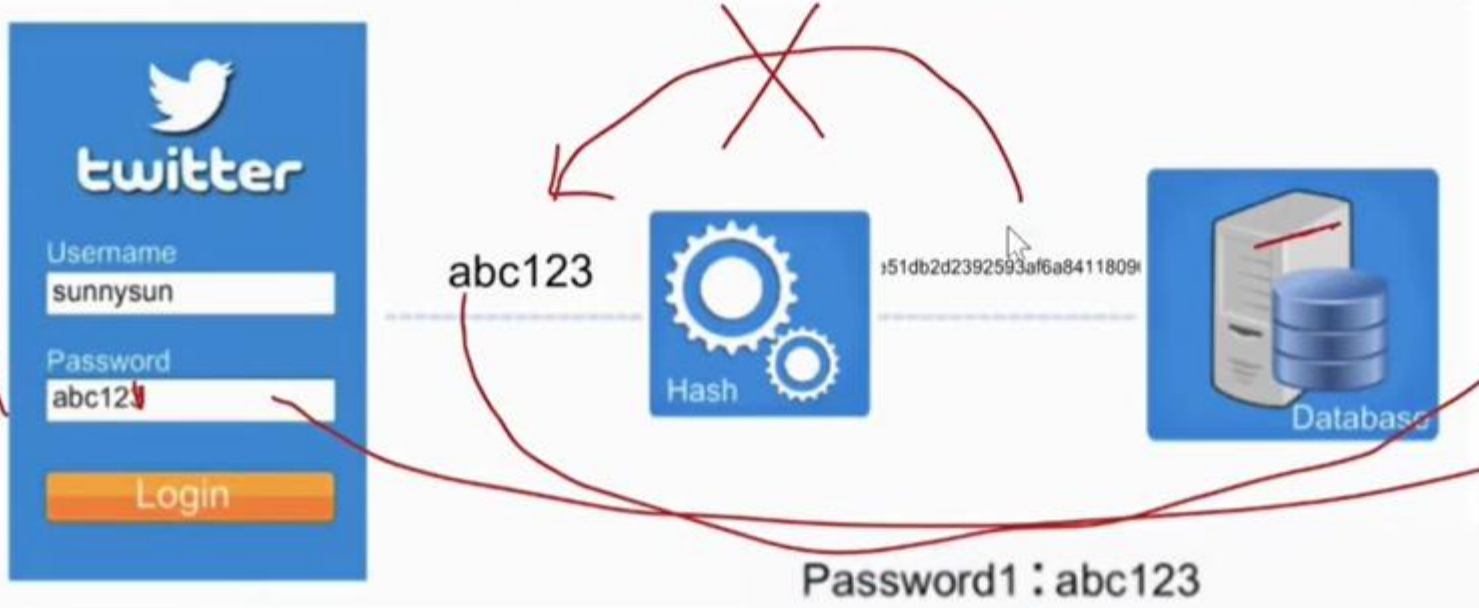
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SHA-256 (Secure Hash Algorithm 256)

HASH ALGORITHM

- **You might say: Hash algorithm is not reversible and our passwords are stored in the form of digests, but why our passwords are still hacked?**

Hashing prevents the original password from being read directly from the database. However, hackers can still 'crack' passwords by guessing common passwords, hashing the guesses, and comparing them to the stolen hash. To protect against this, systems must use strong passwords, unique salts, and slow hashing algorithms like bcrypt or Argon2.